

Recent transformations of land-use and land-cover dynamics across different deforestation frontiers in the Brazilian Amazon



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ABSTRACT

After forest governance reforms by the Brazilian government, Amazon deforestation rates dropped by almost 80% between 2004 and 2012. Since then, however, deforestation has slowly increased again, casting doubts on the long-term sustainability of past conservation policy achievements. Clearly, deforestation rates and the associated local drivers of land-use and land-cover change differ considerably across the region, and adapting public policies to dynamic local contexts and actor constellations remains a major challenge for decision-makers. This paper seeks to contribute to the related science-policy debate by applying frontier theory insights to map, quantify, and compare land-cover dynamics in the Brazilian Amazon between 2004 and 2015. Our theoretical framework performs well in explaining broad variations in scope, nature, and agents of land-use and land-cover change across different frontier regions. We observe two types of transformative processes at deforestation frontiers in the Amazonian context. First, recent frontier development is characterized by an intensification of cattle ranching, and an increasing share of agricultural activities in the production portfolio, which could be the result of better access to modern technologies and markets, combined with forest governance induced scarcity of land for expansion of historically dominant extensive pasture systems. Second, the share of medium and large-scale deforestation declines at first, but rebounds during the observation period in all frontier types after 2012. Further research is needed to better understand the role of the changing political environment in driving these medium and short-term transformations in land-use and land-cover change. Furthermore, policy-makers, international cooperation, and civil society must seek strong integration between value chain based governance measures and effective environmental law enforcement.

1. Introduction

Frontier theory is a prominent conceptual framework to analyze and describe the dynamics of land-use and land-cover change (LUCC) in tropical rain-forest areas (Faris 1999; De Koninck, 2000; Entwisle et al., 2008; Barbier, 2012; Pacheco, 2012). Frontiers can be generally framed as “regions just beyond or at the edge of human settlement” (Merriam-Webster, 2004), where multiple land-conversion processes take place that are characterized by the substitution of natural vegetation with domesticated plants for food and feed production. Deforestation is among the most commonly studied phenomena in the frontier literature, as it is often associated with negative impacts on the global climate, biodiversity, and local as well as regional hydrological cycles (Werth and Avissar, 2002; Fearnside, 2005; Foley et al., 2005; Intergovernmental Panel on Climate Change, 2013).

One strength of frontier theory is the focus on a set of processes and underlying causal relationships that are specific to remote areas with

distinct geographical and social characteristics, for example, in studies on deforestation and environmental change. In the past three decades, multiple theoretical approaches have been proposed to link different causal drivers to frontier development at local scale. This theoretical diversity has become a powerful toolkit to describe LUCC trajectories across different socio-economic settings, for example, in the Amazon region (Fearnside, 2001; Walker et al., 2002; Pacheco, 2005; Aldrich et al., 2006; Jepson, 2006; Caldas et al., 2007; VanWey et al., 2007; Browder et al., 2008; Michalski et al., 2010; Carrero and Fearnside, 2011). Case-study based explanatory richness comes with a desirable conceptual pluralism, but it faces limitations in terms of representativeness and cross-case comparability. Given the technical, financial, and political constraints of policy making at national scale, decision-makers can benefit from spatially explicit landscape and regional scale approaches, when targeting heterogeneous frontier conditions (Sayer et al. 2013). Pacheco (2012), for example, has mapped deforestation frontiers in the Brazilian Amazon describing frontiers in terms of land-

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cover and actor characteristics, using data aggregated at municipality level. He classifies frontiers based on their deforestation levels and identifies priority areas for policy action. Along a gradient of deforestation, however, frontier theory suggests considerable changes in the relative importance of the proximate and underlying causes of deforestation (Angelsen 2007; Walker 2012). Policy design may thus benefit from a theory informed selection of classification variables.

A number of recent studies have, moreover, analyzed policy effects on recent deforestation dynamics at regional scale (Assunção et al., 2012; Arima et al. 2014; Börner et al., 2014; Gibbs et al., 2015a,b). These studies have generally found that the Brazilian government adopted effective policy instruments after Amazon deforestation rates had peaked between 2003 and 2004. However, designing a policy mix that promotes long-term sustainable forest transition pathways requires a more comprehensive understanding of how land-use systems have changed in response to forest governance reform (Barbier et al., 2010).

Hence, this paper aims to inform national and subnational decision makers by (1) developing a theory-based, spatially explicit frontier classification for the Brazilian Amazon, and (2) analyzing frontier-specific land-use change dynamics for the period of 2005 until 2015. For this purpose, we have developed a theoretical framework that builds on key insights from frontier literature, as well as case study evidence from the Brazilian Amazon. Our framework, inspired by Von Thünen's model of agricultural land rents, is presented in Section 2.

Our methodological approach and the construction of a spatial database to map frontier regions and quantify LUCC is presented in Section 3. Results are presented in Section 4, where we identify eight different frontier types, characterized by distinct LUCC dynamics and socio-economic characteristics. Although those frontier characteristics correspond to most of our theoretical expectations, we observe a predominance and persistence of cattle ranching in all frontier types, which deviates from frontier theory predictions. However, our results also indicate a frontier-wide paradigm-shift towards more intensive cattle production and an increase in annual crop production. Section 5 discusses our findings and theoretical shortcomings of our proposed model. Section 6 draws conclusions against the backdrop of the current political and economic context of land-use change in Brazil and beyond.

2. Theoretical background

The conversion of primary forest to cropland or pastures is commonly framed as an investment decision driven by expected returns to alternative uses or future appreciation of land values (Dias et al., 2016). However, as we move from well-established agricultural landscapes to remote frontier regions, expectations related to return on investment are based on very different socio-economic realities. Our conceptual framework assumes that market-distance and accessibility are a key underlying determinant of deforestation, and is therefore based on Johan Heinrich Von Thünen's model of agricultural land rents. In this model, Von Thünen assumes a homogeneous state characterized by a central market, uniform biophysical conditions, and no foreign trade. Transportation costs increase with distance to the market and agricultural goods have a limited durability. This leads to a distribution of land-use classes that follow a pattern of concentric circles with more input-intensive and perishable cultures in the inner circles, and extensive land-uses with less perishable products, such as livestock and wood production, in the outer circles around the market (von Thünen, 1910).

The standard Von Thünen model characterizes the distribution of land-use classes in a steady state, where all land has already been converted and assigned to its most profitable purpose. This contrasts with the dynamic nature of land-use change at forest frontiers, as infrastructure and market access improve over time. In the extended Von Thünen framework presented in Fig. 1, agricultural land-use expands geographically into more remote areas as the frontier develops and urban centers grow over time. The distribution of land-conversion is

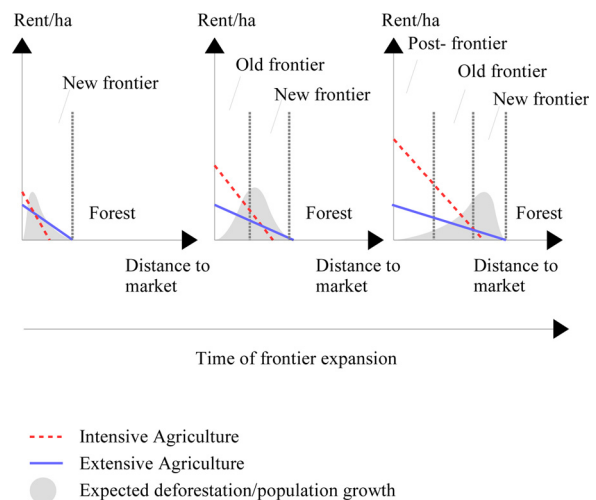


Fig. 1. Population change and deforestation in a time-dependent land-rent model.

dependent on the profitability of different land-use forms, which is dependent on its current distance to the next market and the local market demand (see Fig. 1).

Following Fig. 1, the spatial location of deforestation changes over time as frontier development advances, with the highest deforestation rates occurring in new frontier areas. However, since deforested area accumulates over time, more deforested areas are found in older frontiers compared to newer frontier regions. If new frontiers attract additional settlers, a similar demographic development pattern is expected, with population increases in recent deforestation frontiers and decreasing rural population growth after a certain amount of frontier development. This conjecture is in line with structuralist approaches of frontier theory, although the driving forces behind this development are rooted in economic, as well as demographic changes in local family structures over time (Perz 2003; VanWey et al., 2007; Browder et al., 2008). Consistent with the Von Thünen model, more intensive land-use forms predominate regions that have better access to markets, and frontier development comes along with the replacement of extensive land-use forms with more intensive land-uses - especially in areas close to urban markets.

Despite infrastructure improvements and subsequent population growth, another important factor for frontier development is governance. Policy factors matter, because they can influence land-rents and thereby distort the development of deforestation frontiers (Barbier et al., 2010). In the context of Brazil, planned settlement projects play a crucial role in regional development and land conversion (de Almeida, 1992; Goza, 1994; Moran, 1997; Fearnside, 2008; Pacheco, 2009b). In addition to securing property rights, these projects were thought to come along with infrastructure investments and agricultural extension programs intended to provide settlers with credit, know-how, and production inputs. Despite widespread implementation failures (Pacheco, 2009a), settlement projects were found to promote forest conversion due to higher land rents and policy-induced migration to remote rural areas (Schneider and Peres, 2015). In addition to increased land conversion, we also expect, that the scale of individual deforestation activities is smaller in settlement projects than for other frontier types, because settlers tend to face binding capital and land constraints. Because of these limitations, settlement frontiers are dominated by subsistence-oriented production and/or extensive land-use forms like cattle ranching.

The cultural background of settlers has repeatedly been subject to debates in the literature on Amazon colonization. Farmers, who migrate to the Amazon from the Brazilian South, were shown to prefer different production systems than peasants from the poorer north or north-east of

the country (Pacheco, 2005; Fearnside, 2008; Godar et al., 2012). Culture, capital asset endowment, and region-specific agricultural knowledge thus jointly influenced agricultural practices and associated land-cover dynamics in Amazon settlements. Federal policies attended to the distinct needs of settlers by providing abundant land to poor northeastern peasants as well as entrepreneurial incentives, such as tax reductions and subsidized credits for commercially oriented southern cattle producers. Hence, we expect that land-conversion is more rapid, and production systems are more market-oriented and intense in settlement frontiers dominated by comparatively asset rich South Brazilian migrants.

Another important policy factor in the Brazilian Amazon are protected areas that either ban market-oriented production activities entirely, or strictly regulate the use of and access to land and forest resources. We expect less dynamic population and deforestation patterns in areas with high densities of protected land than in less regulated regions. Besides that, extensive land-use forms are expected to dominate in and around protected areas since agricultural endeavors are more risky (if illegal) and generally more expensive because of higher transportation costs. Furthermore, protected areas are a good indicator for cultural heterogeneity and their corresponding, and often distinct, forest-transition pathways (Perz, 2007). In the Brazilian Amazon, protected areas are commonly established to secure land claims of traditional population groups, thereby creating a space for culture-specific land-use practices that differ from market-driven investment decisions. We also expect a higher presence of subsistence-oriented production forms, including shifting-cultivation, in areas with a dedicated protection status and, correspondingly, a higher share of secondary forest vegetation than in all other frontier regions (van Vliet et al., 2013). Fig. 2 gives a general overview of how we frame frontier successions in this research, and how policy factors might influence time and intensity of primary forest cover removal in pre-frontier areas. In this graph, subsequent waves of frontier succession occur over time and policies influence the time and intensity under which a pre-frontier region is incorporated into the frontier development process.

Previous studies had a working concept of the frontier as a zone of active land conversion from forest to agriculture (De Koninck, 2000; Perz and Walker, 2002; Rudel et al., 2005; Angelsen, 2007; Caldas et al., 2007; Chomitz et al., 2007; Browder et al., 2008). In consideration of the discussion above, we extend this concept and define frontiers generally as *zones of active land conversion with differing degrees of primary forest cover removal and different land-use types that depend on the*

local accessibility, population density, cultural background and governance factors.

3. Data and methods

Our identification and characterization of frontier types in the Brazilian Amazon involves several analytical steps, which are summarized in Fig. 3 and documented on the Sections 3.1–3.3 below.

3.1. Spatial database

Our study region comprises all areas with incidences of clear-cut deforestation in the Brazilian part of the Amazon Biome that were monitored by the Brazilian National Institute for Space Research (INPE) until 2015. We excluded quasi-natural landscapes from our sample in order to produce a frontier classification that appropriately differentiates among anthropogenized zones, as opposed to merely distinguishing forest from non-forest areas. We chose a 1×1 km grid resolution as our unit of analysis to minimize the misleading aggregation of spatially heterogeneous deforestation pathways in larger administrative units, as discussed in Perz (2007).

We have three groups of variables for our analysis (Table 1). The first group comprises the classification variables for the proposed frontier typology based on our methodological framework. The second group contains two variables to sub-classify the settlement cluster according to the origin of settlers. The third group contains variables that were used to descriptively analyze all frontier types in terms of their agro-economic characteristics and LUCC dynamics.

Two travel-time maps were created to measure market accessibility. The first map contains accumulated travel-times to the next municipality center in 2004, irrespective of its population size. The second map contains travel-times to the next large urban market in 2004, which we defined as a city with more than 50,000 inhabitants. We provide a detailed description of the input data and processing steps to create the travel cost maps in the supplementary material of this article (Section 2). For estimating the population density, we used high-resolution population data at the scale of 1×1 km from the Brazilian Institute for Geography and Statistics - IBGE (Supplementary material, Section 3).

Polygon data depicting the location of settlement projects in the Amazon was provided by the Brazilian Institute for Colonization and Agrarian Reform (INCRA 2016). After data cleaning, we converted the polygon data into a binary raster-map (Supplementary material, Section 4). To describe the origin of settlers within a settlement project, two variables were created using IBGE Census data from 2010 (IBGE 2011). A variable called “northernness” indicates whether the settlers originate from the north or south of Brazil and a variable called “easterness” indicates whether they come from the east or west. Both variables can take values between 0 and 1. For the case of “northernness”, 1 represents a population that originates solely from the extreme north, whereas a 0 a population solely from the extreme south of Brazil. In the case of easterness, a 0 value represents population that originates solely from the extreme west, and 1 a population that is solely from the extreme east of Brazil. All values between 0 and 1 are proportional to their geographical distances in the coordinate space. For more information on how these variables were created, please refer to the Supplementary material (Section 5). As with settlements, we created a binary raster map for protected areas that shows if a study cell is either covered by a protected area or not using a composed data-set by Soares-Filho et al. (2010).

3.2. Clustering

The classification of frontier areas consists of a two-step clustering approach. The first step involved classifying the whole data-set based on the six main clustering variables. In the second step, the settlement-

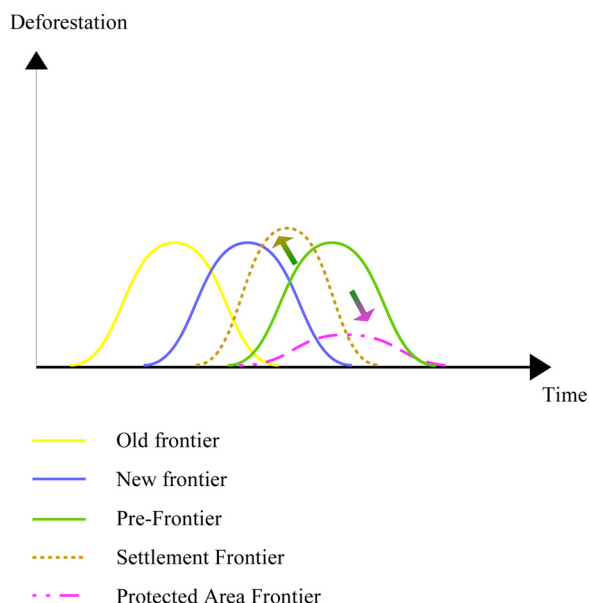


Fig. 2. Frontier Successions and the impact of policies on deforestation.

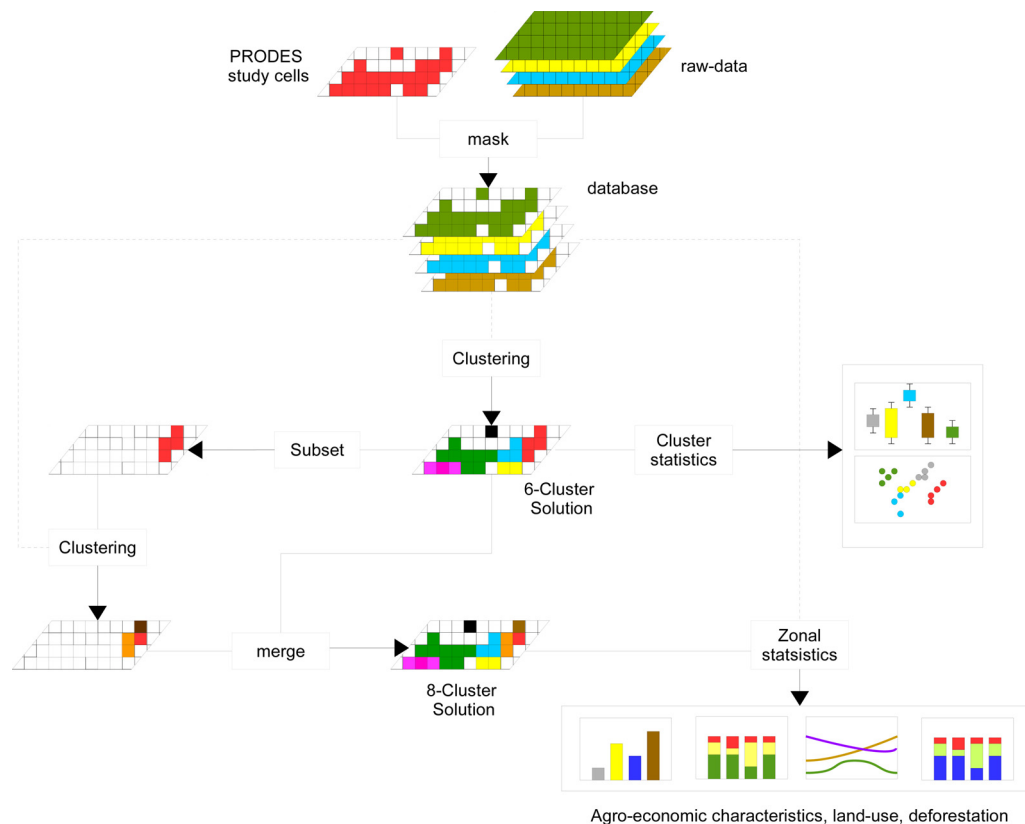


Fig. 3. Overview of the undertaken analysis steps.

cluster from the first cluster-solution was extracted and the observations were reclassified according to their population origins (see Fig. 3).

In both steps, the clustering variables were compiled in a raster-stack for our study area. Extracting those values from the raster yields in a data-set of 1.2 million valid observations. For the classification procedure we chose the *CLARA Program for Clustering Large Applications* (Kaufman and Rousseeuw, 2005), which is available in the *Cluster* package in R (Maechler et al., 2017). The *CLARA* algorithm performs cluster-analyses on several sub-samples of the original data-set and later predicts the cluster-association of all observations based on the best

clustering result. The sub-samples of the current study contained 2.400 observations. After drawing multiple samples from the original data-set, *CLARA* uses the *Partitioning Around Medoids* (PAM) algorithm to find stable structures for a specified number of clusters. To calculate the similarity between clustering-objects we used *euclidean distances* and standardized all variables (z-scores) before applying PAM (details in Section 6 of the Supplementary material).

The binary variables (settlements and protected areas) were treated as if they were numerical variables in the clustering process. This results in a strong influence of these variables on the overall clustering

Table 1

Utilized data-sources.

Variable	Scale	Source	Year
Classification Variables			
Travel time to next municipality centre	continuous	see Supplementary material, Section2	2004
Travel time to next large city (> 50.000)	continuous	see Supplementary material, Section 2	2004
Population density	continuous	Do Carmo Bueno (2016)	2010
Settlements	binary	INCRA (2016)	2004
Protected Areas	binary	Soares-Filho et al. (2010)	2004
Classification Variables – Settlements			
Northernness	continuous	see Supplementary material	2010
Easternness	continuous	see Supplementary material	2010
Descriptive Variables (Agro-economic characteristics & LUCC dynamics)			
Production Inputs per skqm of agricultural area	continuous	IBGE (2009)	2006
Rent obtained from agriculture per skqm of agricultural area	continuous	IBGE (2009)	2006
Salaries paid in agriculture per skqm of agricultural area	continuous	IBGE (2009)	2006
Availability of big tractors (> 100 hp) per skqm of agricultural area	continuous	IBGE (2009)	2006
Availability of small tractors (< 100 hp) per skqm of agricultural area	continuous	IBGE (2009)	2006
Total deforested area	continuous	INPE (2016)	2004–2015
Deforestation patch sizes	continuous	INPE (2016)	2004–2015
Share of agricultural land-cover classes	continuous	INPE (2016)	2004–2014
Stocking rates in Animal Units	continuous	see Supplementary material, section 12	2004–2015

process. This desired effect identifies regions with peculiar governance regimes and corresponding implications for environmental law and land-use policy design.

We compared different cluster solutions with a total amount of 1–10 clusters to determine the optimal number of clusters. This number is based on the average silhouette widths of different cluster solutions (Section 6). We provide a report on the clustering process in the results section with a cluster interpretation based on the variable characteristics of the clusters groups from the chosen classification.

3.3. Measuring agro-economic characteristics and land-cover dynamics inside frontier clusters

The first set of descriptive variables characterizes our frontier clusters in terms of their local agro-economic production conditions in 2006. We included statistics on available machinery (small and big tractors), utilized workforce, as well as acidic soil correctives, fertilizers, and pesticides (the latter three aggregated and labeled as “inputs”). These inputs indicate the degree of agricultural intensification in each cluster that is associated with frontier development as outlined above. To create a data-set that most closely matches our unit of analysis, census-tracts were utilized (IBGE, 2009). Data processing steps are described in the Supplementary material (Section 7). Two frontier clusters in remote areas were excluded from the descriptive analysis due to insufficient numbers of Census observations.

The second set of variables contains information on deforestation dynamics in different frontier clusters. We analyze the sum of deforested areas between 2004 and 2015, as well as size and complexity of deforested patches from the PRODES data-set (INPE, 2016). A short description is given in the Supplementary material as well (Section 8). The PRODES methodology defines forest as dense, open, or mixed ombrophilous forest, and deforestation refers to the complete removal of the forest cover in its primary form (“desmatamento corte raso”). The data does not cover forest degradation due to selective logging, hence the earliest stages of forest conversion are often undetected. INPE also developed a system for monitoring intermediate stages of forest degradation called DEGRAD which started in 2007 but was discontinued in 2013. We did not include DEGRAD data in our analysis because a quantitative assessment of DEGRAD shows an overall low reliability of DEGRAD areas as early indicators for subsequent clear-cut forest cover

removal, with conversion rates ranging only between 1–12% (INPE, 2017). Furthermore, the intermediate spatial resolution of DEGRAD (6.25 ha) is also unable to detect very early stages of forest conversion and selective logging and thus adds little value to our analysis.

The last set of variables describes land-cover trends inside the frontier clusters. To calculate cluster-specific land-cover statistics we use data from the TERRACCLASS project for the years 2004, 2008, 2010, 2012 and 2014 (Coutinho et al., 2013). This study focuses its analysis on the main land-cover types in deforested areas, such as annual agriculture, clean and degraded pastures, secondary vegetation, and mixed land-uses. The latter is used as an indicator for small-scale and subsistence-oriented farming. Perennial cultures are not part of the TERRACCLASS assessment and therefore omitted from our analysis. Summary statistics for the evolution of different land-cover types from 2004 to 2014 were calculated for each cluster, in each year, using the *zonal statistics* function from the raster package in R (Hijmans and van Etten 2012). To characterize the intensity of pasture systems, we look at the differences between the shares of clean and degraded pastures from the TERRACCLASS base-data. Finally, we also include statistics on annual pasture stocking rates measured in animal units per pasture area. This variable was derived from a combination of data-sources described in the Supplementary material of this article (Section 12).

4. Results

4.1. Frontier classification and demographic characteristics of different frontier types

Our clustering variables exhibited sufficient data-agglomerations in terms of accessibility, population density, and governance to discriminate between different types of deforestation frontiers in the Brazilian Amazon. However, formally acceptable silhouette widths ($s > .5$) were obtained only for either four or six frontier clusters (see Fig. B in section 9 in the Supplementary material). Despite the somewhat lower silhouette width, we focus on the six-cluster solution for the purpose of this analysis, because it describes rural areas at a more policy-relevant level of detail.

Fig. 4 represents the cluster-objects from the six cluster solution in 2-dimensional space. The higher dimensional-data is reduced by means of Principal Component Analysis (PCA), with plot dimensions given by

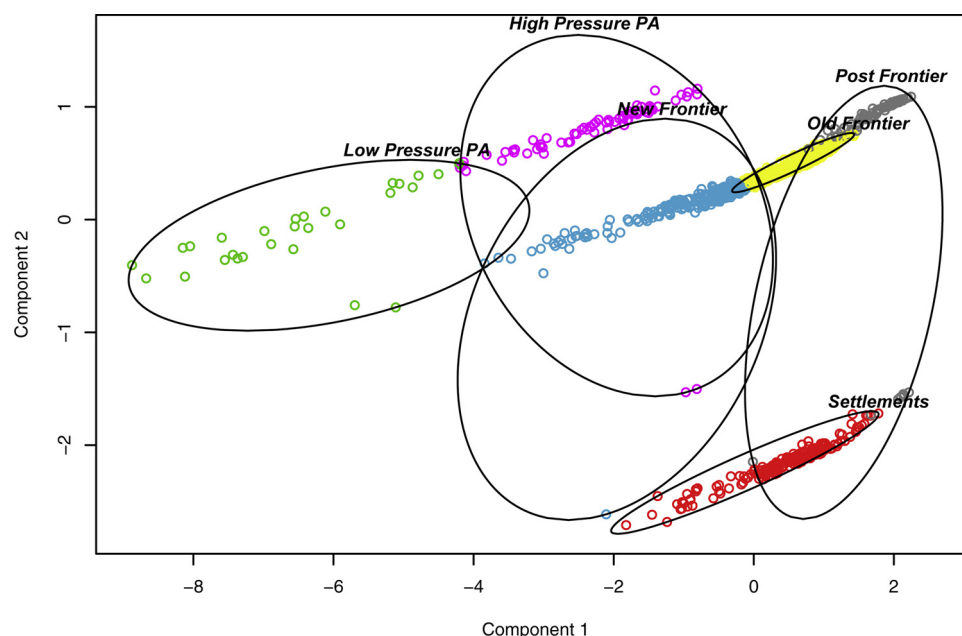


Fig. 4. Clusterplot of the proposed six cluster solution from the first clustering step.

the first two components (Pison et al., 1999). The cluster-objects are visualized as points and the cluster-groups are visualized as ellipsoids around them. In the proposed six cluster solution, the two vectors represent 64% of the point variability. The horizontal axis can be best described as the accessibility and population dimension where population density and market accessibility increase from left to right. The vertical axis depicts the governance dimension where observations cluster in their respective governance categories (protected areas on top, no formal governance status in the middle and settlements at the bottom). Fig. 4 provides several important insights.

First, the governance categories strongly separate the study cells given that they entered the cluster analysis as binary variables. However, there do exist exceptions such as more populated and accessible urban centers in the settlement cluster. Other important exceptions are very remote areas without particular governance status that appear to be most similar to remote protected areas due to their relative isolation and low population density. Note that Fig. 4 shows only the sample of the CLARA algorithm with 1200 study-cells, i.e. extreme values are not necessarily statistical outliers.

Second, the accessibility/population dimension shows smooth transitions between the clusters instead of sharp boundaries. In other words, there is a continuum of frontiers in terms of these cluster dimensions, rather than strictly separable local realities. This is different in protected areas, where accessible and inaccessible regions were clustered into two distinct groups. This finding reflects the fact that some protected areas are designated to protect forests and native communities in regions of very high deforestation pressure, and others are created deliberately in regions of lower commercial interest, where protection is politically less contentious (see A Pfaff et al., 2009).

Sub-clustering was applied exclusively to the settlement cluster and revealed an average silhouette width above 0.6 with $k = 3$. Given that the variables are derived from geographic coordinates, the silhouette width increases with k , as higher numbers of clusters more adequately

represent geographic concentrations of the local populations. The three cluster-solution, nonetheless, best describes the population in broad geographical terms separating origins into the categories north, south, and north-east of Brazil.

The map in Fig. 5 shows the spatial distribution of all frontier types after predicting the cluster associations of all grid-cells. Below we describe each cluster in terms of the cluster variables (for a graphical overview see Fig. C in Section 9 in the Supplementary material).

The *Post Frontier* boasts very short travel times to municipality center and urban markets, as well as high population densities. *Post Frontier* areas can be found across the whole region and are either urbanized or at the rural-urban interface. The *Old Frontier* is comparable to the *Post Frontier* in terms of its accessibility. However, comprised largely by rural areas, the *Old Frontier* has a much lower population density than the *Post Frontier*. *Old Frontiers* engulf *Post Frontier* areas, or main transportation routes, such as the eastern part of the Transamazon Highway and the Amazon river. They cover large areas of eastern Acre, Rondônia, central Mato Grosso, eastern Pará, northern Maranhão, and northern Tocantins.

Areas that are characterized by low accessibility are labeled as the *New Frontier*. They are not only more remote to large urban markets, but also to smaller towns. This is because they rely upon unpaved roads and rivers as main routes of transportation. *New Frontier* areas are typically less densely populated than Old Frontiers. From a geographical perspective, it can be observed that there is often a spatial transition from *Post Frontier*, to *Old Frontier*, to *New Frontier* areas and then to protected areas. This spatial distribution reflects the Von Thünen logic of a decreasing land-use intensity as we move from central markets to remote areas.

We denominate regions inside protected areas that are well accessible as *High Pressure Protected Areas* (short: *High Pressure PA*). They are often found adjacent to the boarder of a protected area and are frequently logged and occupied by agents from other frontiers. Invasions

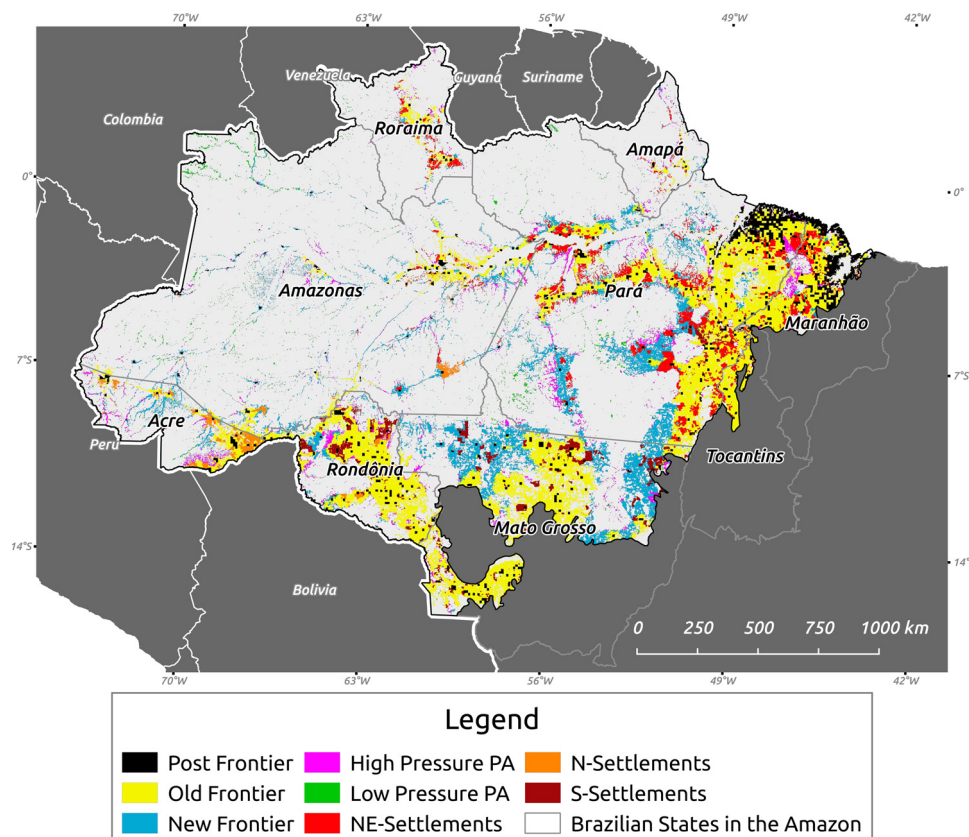


Fig. 5. Map of current deforestation frontiers in the Brazilian Amazon Biom.

are numerous in those areas and a recent report from IMAZON described prominent cases, such as the National Forest Jamanxim, located next to the BR-163 in southern Pará, the National Forest Bom Futuro and the Extractive Reserve Jaci-Paraná in northern Rondonia (Araújo et al., 2017). In some *High Pressure PA* areas, deforestation is more likely caused by reserve dwellers alone or by both internal and external actors. These dynamics can be seen in the extractive Chico Mendes Reserve in Acre, where land has been converted for market-oriented cattle ranching by outsiders, as well as locally based rubber tapping communities (Vadjunec et al., 2009). Furthermore, cases exist where protected areas are highly accessible and forest conversion is part of traditional land-use practices. This is, for example, the case in the Xingu indigenous area, which is surrounded by infrastructure and agricultural land and where the core area is consequently more accessible than in almost all other protected areas. Despite invaded border regions, considerable parts of deforested areas inside the Xingu core region were labeled as a *High Pressure PA* and should therefore be interpreted with caution.

Low Pressure Protected Areas (short *Low Pressure PAs*) are relatively inaccessible protected areas with low population densities. Many *Low Pressure PAs* may not constitute frontiers in the classical sense, because their rapid occupation from outside is rather unlikely in the near future. Most *Low Pressure PA* study-cells lie in the states of Amazonas and Pará.

Settlement frontiers are found in the whole Amazon basin except the extreme Northwest. Most settlers who originated from the south of Brazil occupy settlements in the southern Amazon (geographically closer to their origins), which were denominated as the Southern-Settlement Frontier (short *S-Settlement Frontier*). Migrants from the north-east tend to occupy settlements in the east which were labeled as North-Eastern-Settlement Frontier (short *NE-Settlement Frontier*). Settlers from the north are often found in the west and in the center of the study region. Those areas were denominated as Northern-Settlement Frontier (short *N-Settlement Frontier*). However, this trend is not exclusive. Areas of the *S-Settlement Frontier* can be found in Marabá (eastern Pará), areas of the *N-Settlement Frontier* in central Rondônia, and *NE-Settlement Frontiers* can be found along the BR-163 highway, the Transamazon Highway, and in Roraima.

4.2. Deforestation trends in different frontier regions

A key purpose of our theory-informed classification exercise was to inform policy-makers about how forest loss dynamics coincide with frontier development. It is worth noting again that the overall deforestation trend was decreasing in the study period until 2012, and has since slightly increased. We find that the frontier clusters described above have contributed to the regional trends in ways that are consistent with our theoretical framework following a distinct pattern of frontier successions.

First, the share in total deforestation changes over time according to the frontier status of different regions. Fig. 6 illustrates the share of total deforestation for each frontier type over time including the year 2013 when the *New Frontier* surpasses the *Old Frontier*. Furthermore, the share of deforested areas increases considerably in the *N-Settlement Frontier* and *NE-Settlement Frontier*, and less so in *High Pressure PAs*. The *S-Settlement Frontier* shows less pronounced increases when compared to the other settlement frontiers, whereas the participation of the *Post Frontier* in overall deforestation decreases by a small degree. Recent deforestation increases are, therefore, driven largely by newer frontier areas.

Second, the amount of total deforested areas increases with different stages of frontier development. Fig. 7 shows how much of the original forest cover was lost inside each frontier between 2004 and 2015. Box sizes in Fig. 7 describe the variability of deforestation around the median. As expected, the *Post Frontier* exhibits a high degree of accumulated deforestation, with 50% of its observations ranging between

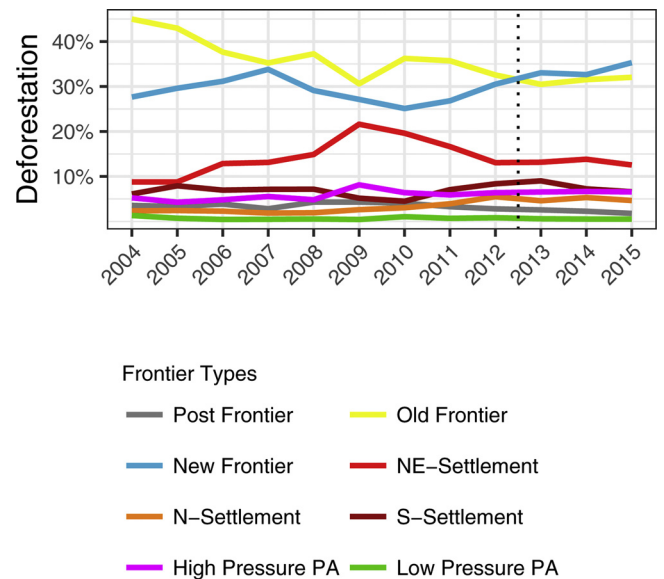


Fig. 6. Participation of different frontier types in total deforestation.

40–90% of total forest cover loss in 2004. The position of the median inside the box (~70%), indicates that more areas are likely to exhibit above-average deforestation levels. Over the observed time period, the interquartile range shrinks, suggesting that a higher level of uniformity is achieved due to rising deforestation levels at the lower bound and relatively stable levels at the upper bound. Upper bound stability points to an advanced stage of forest transition, where no significant forest cover loss is expected to occur in the future. It is worth noting that deforestation levels around the upper bound of the *Post Frontier* are similar to historical forest transition points observed among European countries (Mather, 1992). *Old Frontier* and *Post Frontier* exhibit comparable levels of high overall deforestation, however, the spread of the distribution is larger in 2004, with ~26%–86% inside the box. The median (60%) is also closer to the upper bound and increases over time as the overall spread slightly decreases, indicating more homogeneous, high deforestation levels over time.

The *New Frontier* also meets our expectations with intermediate deforestation levels in 2004 of 8–62% inside the box. Compared to the *Old Frontier*, the *New Frontier* median is closer to the lower bound, and the spread increases until 2015 due to regions with higher deforestation levels. The evolution of deforestation shares suggests an early transition stage in most of the *New Frontier* areas.

High Pressure PAs exhibit higher deforestation levels than *Low Pressure PAs*, however, both have the overall lowest degree of total deforestation. Whereas *Low Pressure PAs* show no significant changes between 2004 and 2015, *High Pressure PAs* exhibit significant increases in the upper bound of the box and an increasing median - from 16% in 2004 to 20% in 2015, i.e. an early stage of frontier development. *N-Settlement Frontier* and *NE-Settlement Frontier* exhibit average levels of accumulated deforestation, but also the most pronounced increases between 2004 and 2015. The spread in both distributions does not change considerably indicating rising levels of forest loss at the lower and no sign of forest transition at the higher end.

The *S-Settlement Frontier* shows, as expected, higher overall conversion rates compared to the other settlement frontiers. However, the scope of forest conversion is much more pronounced than we expected in our theory section, where we assumed that settlements were generally more recent frontier regions. Deforestation trends in the *S-Settlement Frontier* are most comparable to the *Post-Frontier*, and there are converging levels of high deforestation in 2015 for the whole cluster.

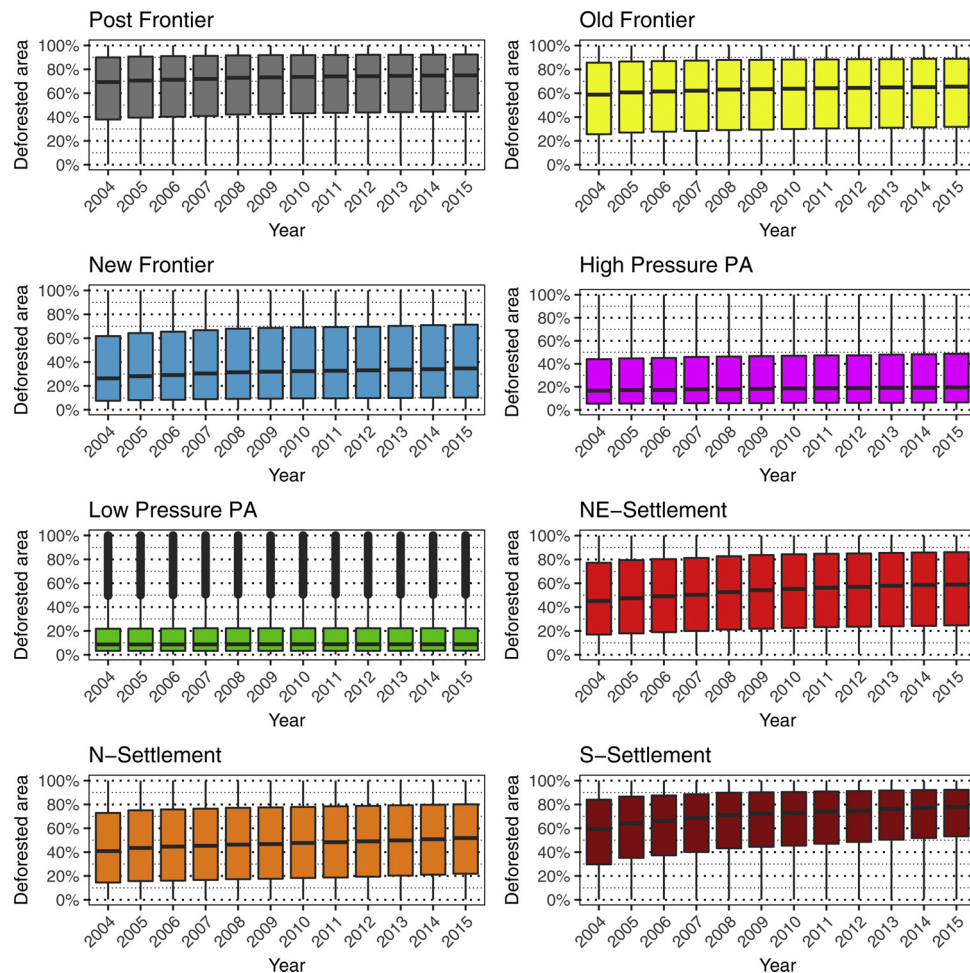


Fig. 7. Total amount of deforested areas in different frontiers.

Beyond overall deforestation trends, the patch-size of deforested areas also differs considerably across frontier types. Fig. 8 shows that small-scale deforestation dominates settlement frontiers and the *Post Frontier*. In contrast, the overall participation of medium and larger patches is higher in the *Old Frontier* and *New Frontier*. However, the contribution of medium and large patches to overall deforestation was unexpectedly high in both *High Pressure PAs* and *Low Pressure PAs*. This is especially the case for the latter, where we would have expected more participation of small-scale deforestation in the overall deforestation composition. However, in the case of illegal territory invasion, the rapid establishment of large land claims after intrusion has been documented in the context of land speculation (Fearnside, 2008). On the other hand, larger patches in these frontiers could also be cumulative agglomerations of small patches which were detected after several years due to cloud coverage, and could potentially be classified as a connected patch (for a discussion of these classification issues see the Supplementary material, Section 11).

Beyond the aforementioned differences, we also observe changes in deforestation behavior across frontier types. Our data suggests a decreasing trend in deforestation patch-size until 2010, which is comparable to other studies (Rosa et al., 2012). Between 2010 and 2015, a few years after the implementation of new forest law enforcement mechanisms in the Brazilian Amazon, the share of medium and large-scale deforestation increased again almost to 2004 levels (see Fig. 8 for cluster-wise comparisons and, for basin-wide statistics, Fig. D in section 11 of the supplementary material).

4.3. Land-use change and agro-economic characteristics in different frontier types

In line with our theoretical expectations, intensive land-uses in the form of annual agriculture show a higher presence in frontiers with good market access compared to more remote areas. Fig. 9 gives a graphical overview of LUCC trends inside different frontier types between 2004 and 2014. Annual agriculture in 2004 is highest in the *Old Frontier* (4.5%) followed by the *New Frontier* (2.3%), the *Post Frontier* (2.1%), and *S-Settlement Frontier* (1.1%). Protected areas, *N-Settlement Frontier* and *NE-Settlement Frontier* exhibit practically no relevant annual agriculture in 2004 (see also Supplementary material, table A in Section 10).

Mixed land-uses have a higher share in the two protected areas and in the *N-Settlement Frontier* and *NE-Settlement Frontier*, which is a sign of small-scale and subsistence oriented agriculture. Furthermore, the share of degraded pastures in those regions is comparably high, indicating lower degrees of pasture management and intensification. As expected, *Low Pressure PAs* are dominated by secondary vegetation, which can be linked to patterns of traditional shifting-cultivation.

The land-cover data also confirms our prediction that frontier development coincides with the increase of more intensive land-use forms and/or the substitution of extensive land-uses by more intensive ones. As expected, increases of annual agriculture are high in the *Old Frontier* (+3.4%), *Post Frontier* (+1.1%), and *S-Settlement Frontier* (+3.3%), and land-use transition is much slower in the *N-Settlement Frontier* and

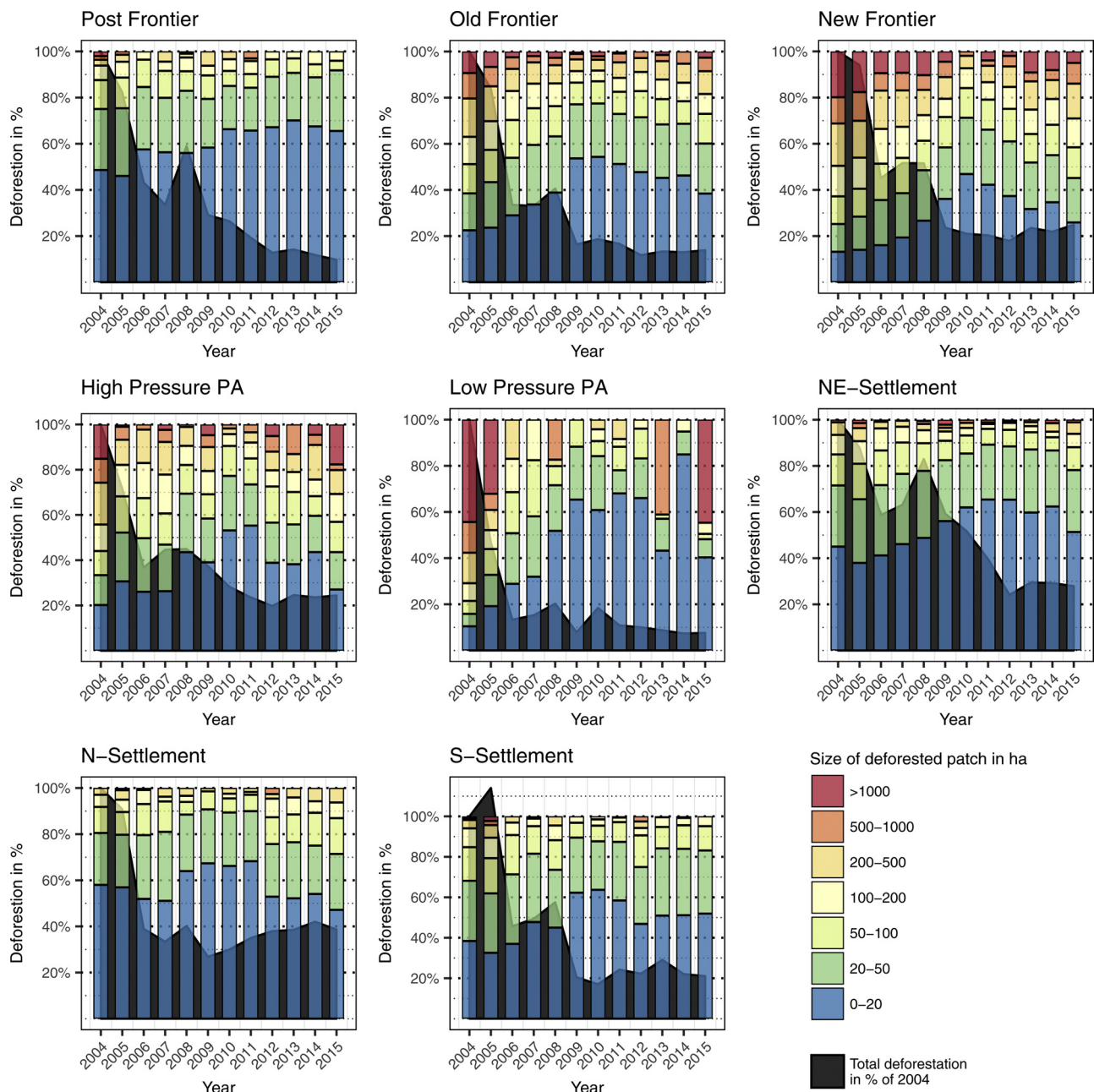


Fig. 8. Patch sizes of clear-cut deforested areas in different frontier types.

NE-Settlement Frontier. Unexpectedly, the *New Frontier* exhibited the second highest share in annual agriculture already in 2004 and, due to high growth rates between 2004 and 2014 (+ 5,8%), boasted the highest share of annual agriculture in 2014 (8,1%). Considering these figures compared to 8,0% annual agriculture in 2014 in the *Old Frontier*, and only 3,19% in the *Post Frontier*, this characteristic is probably driven by the strong growth of soybean agriculture in some of these frontiers and a trend in the integration of cattle and crop production systems. Also, the *Post Frontier* exhibits a high share of mixed land-uses (7,2%), which could indicate that the market demand for the local consumption of input intensive products (fruits and vegetables) is mostly met by small-scale agriculture close to the urban markets as predicted by the Von Thünen model.

Furthermore, degraded pastures decrease in practically all frontier types, which could either be associated with pasture reform, or land abandonment and subsequent forest regrowth. The total share of pastures (clean pastures and degraded pastures) dropped in almost all

frontier types at rates between 5 and 10%, which is different from results obtained for previous study periods (Pacheco, 2012). Given the well-documented dominance of pastures among post-deforestation land uses in the Brazilian Amazon, annual agriculture remains under-represented in the land use mix across frontiers (Andersen, 1996; Pfaff, 1999; Barreto and Silva, 2010; Bowman et al., 2012; Pacheco and Pocard-Chapuis, 2012). This is not necessarily contradicting the Von Thünen logic, because our data base does not distinguish between all types of agricultural production and does not capture transformations that might occur over several decades to come.

Despite the substitution of degraded pastures with clean pastures, we do nonetheless observe an increase in stocking rates in the whole study area irrespective of the frontier type. Fig. 10 illustrates that stocking rates increased on average from 1.3 Animal Units in 2003, to 1.7 Animal Units in 2015. As expected, stocking rates are the lowest in *Low Pressure PAs* and *New Frontier* regions, and higher in *Old Frontiers* and *Post Frontiers*. Interestingly, the *S-Settlement Frontier* and *NE-*

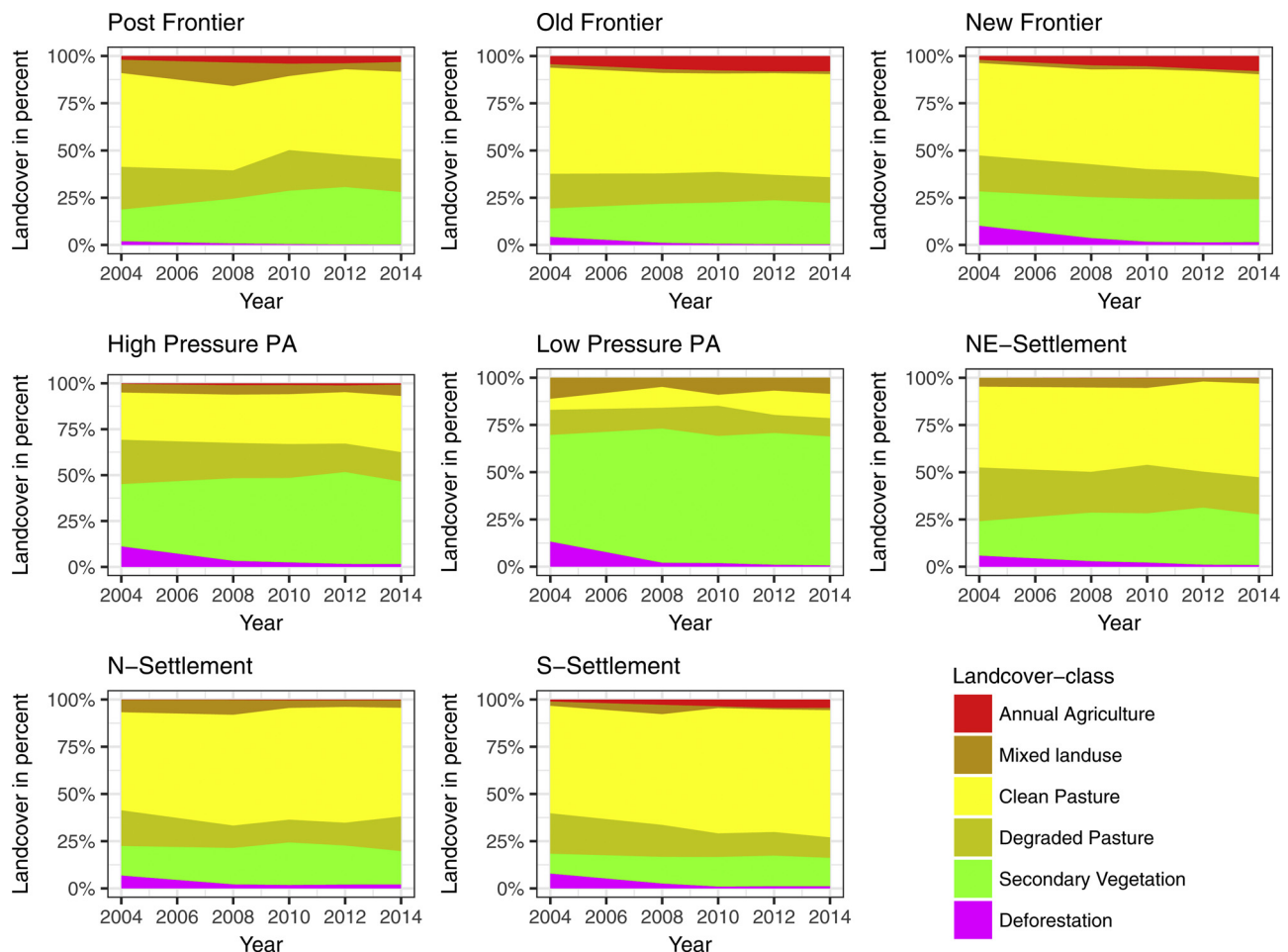


Fig. 9. Land-cover change in different frontier types.

Settlement Frontier exhibit the highest stocking rates suggesting either newer and more productive pastures or overgrazing. The latter is typically associated with pasture degradation and expansion, which might explain persistently high deforestation rates in those areas.

As expected, intensity and pace of land-use transformation are also associated with different degrees of capitalization, agricultural input usage, and obtained rents amongst frontier types. Fig. 11 gives a general overview of the agro-economic characteristics measured in each frontier in 2006, excluding two remote frontiers with insufficient numbers of observations. *Post Frontier* and *Old Frontier* exhibit higher densities of agricultural machinery with 7–8 big tractors per km² of agricultural land versus roughly 4 tractors in other frontier regions.

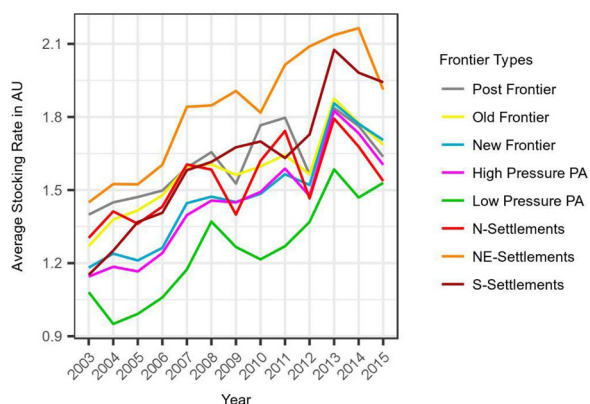


Fig. 10. Stocking Rates amongst different frontier types.

Farmers in these zones rely more heavily on agricultural inputs, such as fertilizers, pesticides, and farm labor (measured in terms of per hectare hired labor costs). In the *Post Frontier*, the highest land rent was obtained per hectare (325 USD or 1150 BRL on average)¹, which is in line with our expectations from the land-rent model. The *New Frontier* was comparatively less capitalized in terms of machinery, but was more dependent on hired labor - which is a requirement in annual agriculture. The second highest per hectare rents were obtained from agriculture (265 USD or 925 BRL per hectare on average), which can also be associated with the high share of annual agriculture in the *New Frontiers'* land-use portfolio. *N-Settlement Frontier* and *NE-Settlement Frontier* relied less on machines, hired labor, and other agricultural inputs. Land rents obtained from agricultural activities were intermediate in those regions in 2006. A unique pattern is found for the *S-Settlement Frontier* where despite higher production inputs, very low per hectare rents were obtained from agriculture in 2006 (only around 104 USD or 363 BRL per hectare). Associated with lower levels of mechanization levels and paid workforce this finding probably reflects the strong predominance of cattle ranching activities at these frontiers.

In conclusion, land-use systems are consistent with our characterization of land-cover dynamics in the frontiers. Settlement Frontiers are dominated by smaller enterprises and family agriculture and a general persistence in cattle production. In contrast, the *Old Frontier* and *New Frontier* have a higher share of large-scale, capital-endowed agriculturists with increasing shares of annual crop production. Protected areas might be subject to land-speculation with large deforestation

¹ Monetary values were deflated and converted to USD in October 2017.

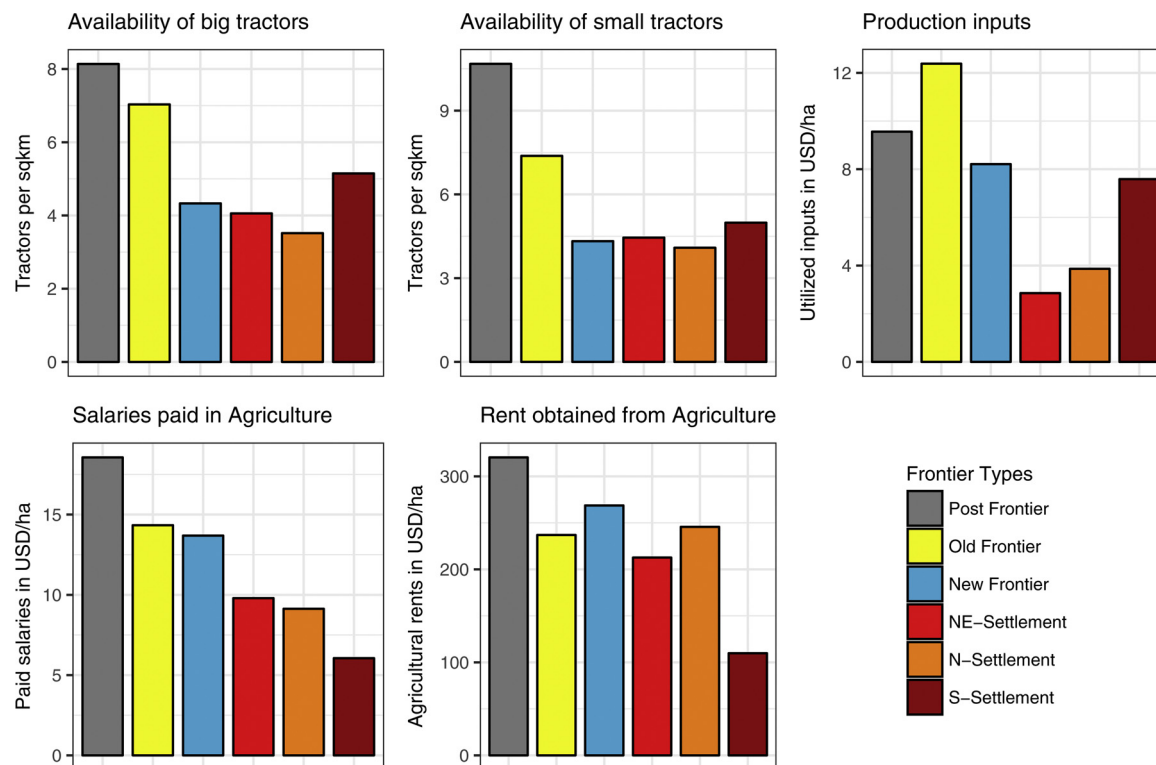


Fig. 11. Agro-Economic Characteristics of different frontier types.

patterns and pasture establishment to corroborate land claims as well as patterns of shifting-cultivation characterized by secondary vegetation.

5. Discussion

Based on our theoretical framework that incorporates site-specific factors and integrates the land-rent model into frontier theory, we identify eight different frontier development types in the Brazilian Amazon using a spatially explicit and data driven classification procedure of anthropogenized landscapes. Our results confirm most, but not all, theoretical expectations with respect to policy-relevant outcome indicators, such as deforestation and agricultural intensification. First and exceptions aside, the land use mix follows an expected gradient of intensive (older frontier regions) to extensive (newer frontiers) land use. This gradient is moderate due to the predominance of cattle production in all frontier types. Contrary to previous studies finding cattle system expansion in all frontier stages (Pacheco, 2012), we observed a gradual substitution of extensive land-use forms with more intensive land-uses in almost all frontiers. Given that our study horizon covers a historical and well-documented forest governance reform in Brazil, this result may suggest an environmental governance-induced paradigm shift in modern frontier development in the Brazilian Amazon. While our focus was not on the mechanisms and impacts of this governance reform, we believe that our findings were modulated by two pivotal policy events in our study period: the 2004 Plan for Protection and Control of Deforestation in the Amazon (PPCDAm) and the reformulation of the Brazilian Forest Code in 2012.

Aided by improved monitoring-technology, the PPCDAm leveraged existing and command-and-control policies to deter illegal deforestation (MMA, 2013). This led to more fines for illegal deforestation and area embargoes (from 2008 onwards) in areas with high deforestation pressure. These measures can be more effective in old frontier areas with transparent land tenure regimes, where the economic consequences of law enforcement and legal coercion are immanent and accessibility is improved. Conditions at these frontiers are conducive to the farm level responses we observe, such as diversification and the

substitution of land-consuming cattle-ranching by annual agriculture. Better law-enforcement also represents a plausible explanation for the increasing share of cleaned pastures in all frontier areas, because reforming degraded areas reduces the demand for new land from deforestation.

Additionally, law enforcement prioritized larger deforestation areas (> 100 ha), which would explain the decreasing occurrence of large deforestation patches in older frontier regions and the persistence of small-scale deforestation in settlement frontiers up to 2012 (Börner et al., 2015). Finally, the new Forest Code's much debated amnesty for land that was illegally deforested before 2008 may have partially watered down previous conservation commitments, but it also created incentives for legalized farms to convert degraded areas into clean, productive pastures in any of the analyzed frontier types. Further research should analyze these mechanisms in more detail.

Despite a general paradigm shift, the pace and nature of agricultural intensification depends crucially on the frontier type. First, while annual agriculture increases mostly in the *Post Frontier*, *Old Frontier*, *New Frontier*, and *S-Settlement Frontier*, pasture related intensification (measured as an increase of clean pastures against degraded ones and a general increase in stocking rates) dominates all frontier types and protected areas. Simultaneously, secondary vegetation increases in all frontiers, which could indicate a land sparing effect of intensification decisions. Second, newer frontiers still experience more deforestation than older frontiers, and the scale (average patch size) of deforestation has been temporarily reduced after forest governance reforms in 2004. We find, however, that small-holder dominated settlement frontiers have remained largely unaffected by this trend, which explains the observed increase in the overall share of small-scale deforestation until 2010. Additionally, our study found that patch size distribution has largely returned to pre-2004 levels in all frontiers after 2010. This finding contrasts with recent studies who associated post-2004 deforestation trends with small-scale farming (Rosa et al., 2012; Godar et al., 2015; Assunção et al., 2016). Furthermore, it suggests that large-and medium-sized deforestation may have only been temporarily put on hold by actors expecting a return to laissez-faire forest governance, or

to invest in less scrutinized Brazilian biomes, such as the Cerrado (Richards et al., 2016).

This paper's main contribution lies in making use of increasingly rich and publicly available spatial and non-spatial data sources to reconcile a macro-scale perspective on land-cover change, with spatial heterogeneity in local constellations of land-use drivers. As such, our approach may also prove useful to inform conservation policy design at other tropical forest margins. A couple of theoretical and methodological caveats remain, nonetheless, that should be addressed in future research.

Our frontier classification is more reliable for older frontier types than for dynamic new frontiers. In future analyses, it would be desirable to concentrate on newer frontiers with an additional set of variables that discriminate better between local differences within emerging frontier areas. We experimented with spatial information on large hydroelectric dams and mining projects as frontier classifiers, but the spatial distribution of these variables reduced the stability of cluster solutions. Selected additional clustering variables could be used to relax strong assumptions made in the Von Thünen inspired frontier framework, such as the uniformity of agro-ecological conditions. Differences in agricultural suitability can explain the persistence of extensive land-use forms in regions that are not suited for intensive agriculture, even if they are close to an urban market. The same applies to the expansion of input intensive activities in suitable, but remote frontier areas. Furthermore, foreign trade effects could be included by measuring distance to export infrastructure, such as processing facilities and ports, to account for global market forces that can affect frontier development. And finally, land rent theory may be insufficient to explain the persistence of pastures as dominant land use forms across frontier types. Beyond economic factors, cattle ranching fulfills social (Garrett et al., 2017) and cultural functions (Hoelle, 2015) that yet need to be incorporated into modern frontier theory.

6. Conclusions

Reconciling environmental conservation and rural development at deforestation frontiers in the tropics requires a mix of conservation and rural development policies that appropriately address heterogeneous frontier development stages and coordinate policy action among historically separate governance domains. The publicly available frontier map developed in this study can support the design of such policy mixes, for example, via analyses of policy impacts and the identification of priority regions for government action (available at Schielein, 2018). Our findings suggest that higher levels of deterrence could stabilize forest loss in the *New Frontier*, where deforestation is on the rise and native vegetation reserves are still sizable. Furthermore, policy-makers should be aware that the composition of forces that drive frontier development at *New Frontiers* changes over time. Agricultural intensification is promoted in the literature as a possible solution to limit land expansion (Cohn et al., 2014; Strassburg et al., 2014), but only if associated with improved environmental governance (Ceddia et al., 2014). Otherwise, government-supported intensification programs such as the Brazilian Development Bank's Inovagro may stimulate rather than alleviate land demand in new frontier areas.

In capital and labor scarce settlement frontiers and High Pressure PAs land expansion is still strongly associated with cattle-dependent small-scale farming. The reasons for this phenomenon are well-studied and include, among socio-cultural aspects, a mistrust in bank savings, safety-net functions, such as liquidity in case of emergencies, low investment costs and low labor intensity, governmental support, and pasture creation as a mechanism to secure land claims (Gomes and Valério, 2009; Vadjunec et al., 2009; Dávalos et al., 2014; Garrett et al., 2017).

Nonetheless, relaxing constraints to agricultural development by improving transport networks and technical extension services could stimulate the substitution of land-consuming cattle activities by market-

oriented annual or perennial agriculture with positive welfare impacts. Such measures will, however, only produce the desired land sparing effect if accompanied by effective environmental law enforcement to discourage extensive land uses that rely on illegal deforestation. Policy mixes along these lines could pay off most in settlements dominated by north-eastern and northern populations, where forest reserves are still abundant and deforestation rates are rising.

Detering deforestation in protected area frontiers should remain among the top priorities for government action, for example, under the PPCDAM. Our frontier map identifies large areas of invasions in the *High Pressure PA* zone where stronger law enforcement is required. In contrast, the trend towards degazettement of protected areas adopted by the Brazilian government in recent years (Bernard et al., 2014) might have already created additional incentives for protected area invasions in Pará and other regions within the biome (Pires, 2017).

Where protected area status allows sustainable land and natural resource use, deforestation pressure could be reduced by strengthening local economic activities that depend on sustainable timber and non-timber forest product extraction. Existing policies such as the National Plan for the Promotion of Sociobiodiversity Value Chains (PNPSB) should specifically target such high risk areas, for example, via effective minimum prices or subsidized credit and insurance schemes.

International cooperation and NGOs increasingly promote specific value chain based governance measures (Lambin et al., 2014). However, to the extent that our Von Thünen inspired framework explains forest conversion dynamics in the Brazilian Amazon, commodity-based value chain governance can only partially influence land use decisions at the frontier and remains vulnerable to cross-frontier leakage and spillover effects. Our results thus emphasize the important role of effective environmental law enforcement as a crucial backbone for conservation success across the diverse frontier landscapes in the region.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.landusepol.2018.04.052>.

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